

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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Date:

1. TITLE

Develop a Responsive Webpage Using HTML5 and CSS3

2. AIM

To develop a responsive Student Learning Portal using HTML5 and CSS3 styling with flexbox layouts, hover effects, CSS gradients, rounded borders, and media queries.

3. OBJECTIVE

The objectives of this experiment are:

- To understand CSS3 selectors, box model properties, and color specifications.
 - To implement modern CSS linear gradients for header and welcome cards.
 - To create flexible multi-column card layouts using CSS Flexbox (display: flex).
 - To apply CSS hover animations (transform: scale, transition) for interactive UI elements.
 - To style navigation bars, custom buttons, faculty cards, and footer sections.
 - To implement media queries (@media) for responsive mobile screen adaptation.
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4. THEORY

CSS3 (Cascading Style Sheets) enhances HTML elements with layout presentation, vibrant visual aesthetics, and responsive layout behavior across devices.

Key CSS3 features used in this Student Learning Portal application:

- Flexbox (display: flex) creates flexible row and column card containers.
- Linear Gradients (linear-gradient) generate vibrant color background headers.
- Box Shadows & Rounded Corners (box-shadow, border-radius) add visual depth.
- Hover Transitions (:hover, transform: scale) elevate user interactivity.
- Media Queries (@media (max-width: 700px)) adjust layout dynamically for mobile viewports.

The basic flow of the application is:

HTML Document + CSS Styling → CSS Box Model & Flex Layout Parsing → Responsive Browser Render

5. ALGORITHM / PROCEDURE

1. Create exp2.html and exp2.css files.
 2. Link exp2.css inside exp2.html using <link rel="stylesheet" href="style.css">.
 3. Structure the HTML document with <header>, <nav>, <main>, <section>, <article>, and <footer>.
 4. In CSS, set global box-sizing: border-box and default font family.
 5. Style <header> using a linear-gradient background, white centered text, and custom font sizes.
 6. Style <nav> with a dark background (#17202a) and interactive link hover effects (color: #f1c40f).
 7. Build a welcome banner section styled with vibrant orange-red linear gradient.
 8. Create a course container using display: flex and gap: 20px for multi-column alignment.
 9. Create three course cards (.python, .web, .ai) with custom background colors and rounded corners.
 10. Add hover animation effect (.course:hover { transform: scale(1.05); transition: 0.3s; }).
 11. Style faculty cards with left accent borders (border-left: 6px solid #1abc9c).
 12. Style the contact section and footer with dark matching themes.
 13. Add a media query (@media (max-width: 700px)) to stack course cards vertically on mobile screens.
 14. Open exp2.html in a web browser to verify visual design.
 15. Perform responsive screen test cases and verify hover interactions.
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6. PROGRAM

exp2.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Learning Portal</title>
  <link rel="stylesheet" href="exp2.css">
</head>

<body>

  <header>
```

```
<h1>Student Learning Portal</h1>
<p>Learn Today, Lead Tomorrow</p>
</header>

<nav>
  <a href="#home">Home</a>
  <a href="#courses">Courses</a>
  <a href="#faculty">Faculty</a>
  <a href="#contact">Contact</a>
</nav>

<main>

  <section id="home" class="welcome">
    <h2>Welcome Students!</h2>
    <p>
      Explore courses, meet our faculty and improve
      your technical skills.
    </p>
    <button>Explore Now</button>
  </section>

  <section id="courses">
    <h2>Our Courses</h2>

    <div class="course-container">

      <article class="course python">
        <h3>Python Programming</h3>
        <p>Learn Python programming from basics to advanced.</p>
      </article>

      <article class="course web">
        <h3>Web Development</h3>
        <p>Learn HTML, CSS and JavaScript to build websites.</p>
      </article>

      <article class="course ai">
        <h3>Artificial Intelligence</h3>
        <p>Explore AI, Machine Learning and modern technologies.</p>
      </article>

    </div>
  </section>

  <section id="faculty">
    <h2>Our Faculty</h2>

    <article class="faculty-card">
      <h3>Dr. Ramesh Kumar</h3>
      <p>Professor - Computer Science</p>
    </article>

    <article class="faculty-card">
      <h3>Mrs. Priya Sharma</h3>
      <p>Assistant Professor - Information Technology</p>
    </article>
  </section>
</main>
```

```

        </article>
    </section>

    <section id="contact">
        <h2>Contact Us</h2>
        <p>Email: studentportal@gmail.com</p>
        <p>Phone: 9876543210</p>
        <p>Address: Visakhapatnam, Andhra Pradesh</p>
    </section>

</main>

<footer>
    <p>&copy; 2026 Student Learning Portal | All Rights Reserved</p>
</footer>

</body>
</html>

```

exp2.css

```

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: Arial, sans-serif;
    background-color: #f5f7ff;
    color: #333;
}

/* Header */
header {
    background: linear-gradient(135deg, #6a11cb, #2575fc);
    color: white;
    text-align: center;
    padding: 40px;
}

header h1 {
    font-size: 38px;
    margin-bottom: 10px;
}

header p {
    font-size: 18px;
}

/* Navigation */
nav {
    background-color: #17202a;
    padding: 18px;
}

```

```
        text-align: center;
    }

    nav a {
        color: white;
        text-decoration: none;
        margin: 0 25px;
        font-size: 17px;
        font-weight: bold;
    }

    nav a:hover {
        color: #f1c40f;
    }

    /* Main */
    main {
        width: 85%;
        margin: 30px auto;
    }

    /* Sections */
    section {
        background-color: white;
        padding: 30px;
        margin-bottom: 25px;
        border-radius: 12px;
        box-shadow: 0 4px 10px #d5d8dc;
    }

    section h2 {
        text-align: center;
        color: #6a11cb;
        margin-bottom: 20px;
    }

    /* Welcome Section */
    .welcome {
        background: linear-gradient(135deg, #ff9966, #ff5e62);
        color: white;
        text-align: center;
    }

    .welcome h2 {
        color: white;
        font-size: 30px;
    }

    .welcome p {
        margin-bottom: 20px;
        font-size: 18px;
    }

    button {
        background-color: #2ecc71;
        color: white;
```

```
    border: none;
    padding: 12px 25px;
    border-radius: 25px;
    font-size: 16px;
    cursor: pointer;
}

button:hover {
    background-color: #27ae60;
}

/* Course Cards */
.course-container {
    display: flex;
    gap: 20px;
    justify-content: center;
}

.course {
    flex: 1;
    padding: 25px;
    color: white;
    border-radius: 10px;
    text-align: center;
}

.course h3 {
    margin-bottom: 12px;
}

.python {
    background-color: #3498db;
}

.web {
    background-color: #e67e22;
}

.ai {
    background-color: #9b59b6;
}

.course:hover {
    transform: scale(1.05);
    transition: 0.3s;
}

/* Faculty */
.faculty-card {
    background-color: #e8f8f5;
    padding: 20px;
    margin: 15px 0;
    border-left: 6px solid #1abc9c;
    border-radius: 5px;
}
```

```

.faculty-card h3 {
    color: #16a085;
    margin-bottom: 8px;
}

/* Contact */
#contact {
    background-color: #fff3cd;
}

#contact h2 {
    color: #d68910;
}

#contact p {
    margin: 10px 0;
}

/* Footer */
footer {
    background-color: #17202a;
    color: white;
    text-align: center;
    padding: 20px;
}

/* Responsive Design */
@media (max-width: 700px) {

    main {
        width: 95%;
    }

    .course-container {
        flex-direction: column;
    }

    nav a {
        display: block;
        margin: 12px;
    }
}

```

7. CODE EXPLANATION

Code	Explanation
linear-gradient()	Creates smooth color transitions for background headers and banners.

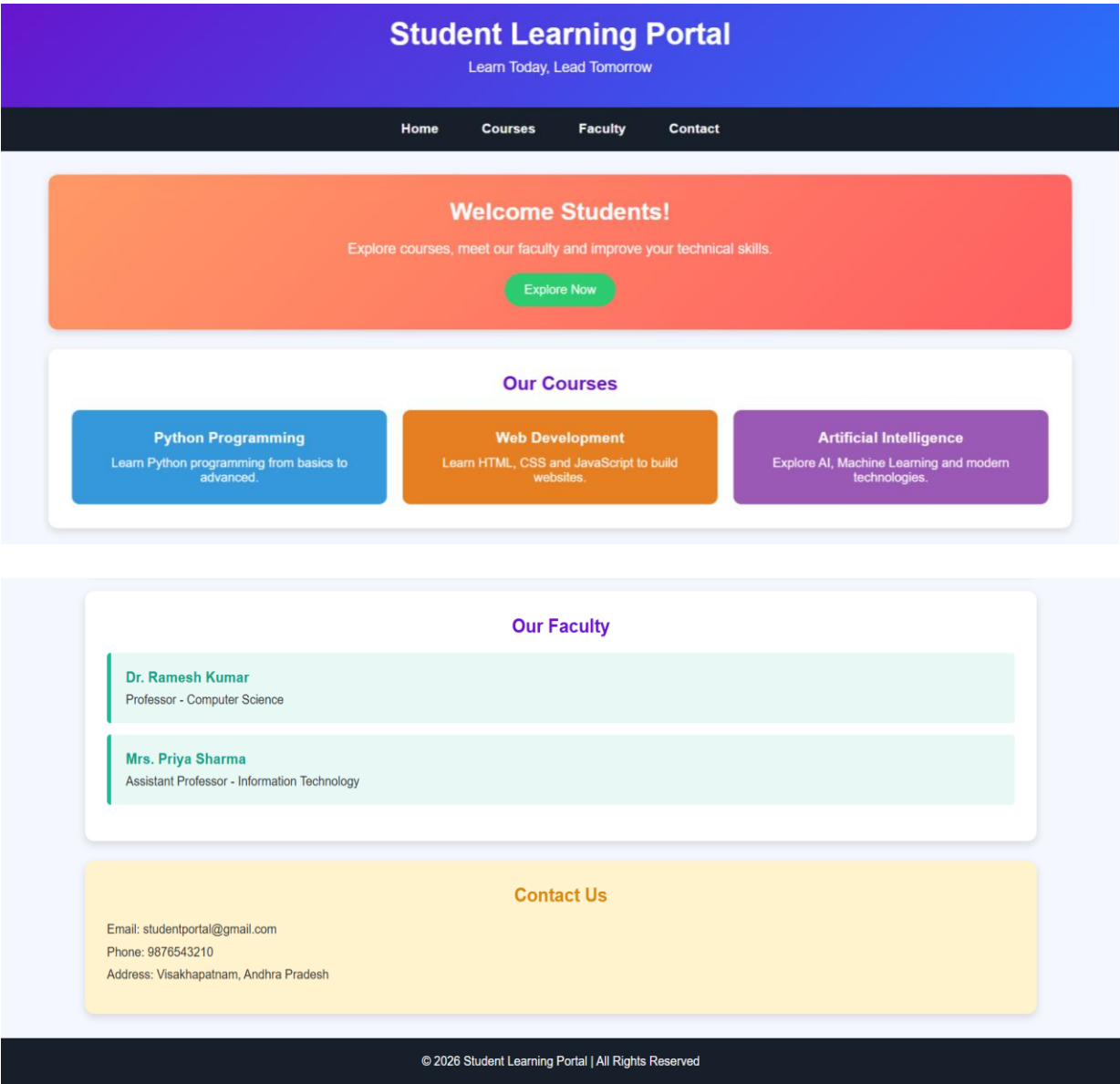
display: flex	Enables flexbox layout mode for course card containers.
gap: 20px	Sets consistent spacing between course cards.
border-radius	Rounds card corners for a modern, sleek UI appearance.
box-shadow	Applies shadow drop effects to section containers for visual depth.
:hover	Applies dynamic styles when mouse hovers over navigation links and buttons.
transform: scale()	Enlarges course cards smoothly during hover events.
@media (max-width: 700px)	Applies responsive styles when screen width is 700px or smaller.

8. TEST CASES AND EXECUTION DATA

Test Case	Feature Tested	Test Input	Expected Result	Actual Result	Status
TC-01	Navigation Link Hover	Hover over Home/Courses	Link turns yellow (#f1c40f)	Link turned yellow	Pass
TC-02	Course Card Hover	Hover over Python card	Card scales by 1.05x	Card scaled smoothly	Pass
TC-03	Button Hover Effect	Hover over Explore button	Button darkens to green	Button color changed	Pass
TC-04	Faculty Card Border	View Faculty section	Green left accent border	Accent border rendered	Pass
TC-05	Mobile Responsive View	Resize screen < 700px	Flex cards stack vertically	Cards stacked in single column	Pass

9. OUTPUT

Output 1: Student Learning Portal Desktop Webpage Execution



10. RESULT

Thus, a responsive Student Learning Portal webpage was successfully developed using HTML5 and CSS3. Flexbox layouts, hover effects, linear gradients, and responsive breakpoints were verified across multiple test cases.